BACKING AND INITIALISING STORE CONTROL UNIT REGISTERS

BIT	COMMAND	STATUS
0	INHIBIT DATA CHECK	DATA CHECK INHIBITED
1	INHIBIT ROUTINING	ROUTINING INHIBITED
2	INVERT BUFFER PARITY	BUFFER PARITY INVERTED
3	INHIBIT FREEZE	FREEZE INHIBITED
4	SET INITIALISING TWO	INITIALISING TWO
5	SET INITIALISING ONE	INITIALISING ONE
6	INHIBIT DISC NOT ROTATING	DISC NOT ROTATING INHIBITED
7	SPARE	SPARE
8	"	"
9	NON FREEZE FAULT RESET	INTERNAL PARITY FAULT
10	FORCE READ DATA TOGGLE	WRITE PARITY FAULT
11	SPARE	DATA BUFFER OVERFLOW
12	DIAGNOSTIC CLOCK MODE	SECTORS PER TRACK FAULT
13	FORCE CLOCK PULSES ZERO	WORDS PER SECTOR FAULT
14	FORCE CLOCK PULSES ONE	TRANSFER TIMEOUT
15	FORCE CLOCK PULSES TWO	INVALID TRANSFER CAPABILITY
16	FORCE CLOCK PULSES THREE	DATA CHECK FAULT
17	FORCE CLOCK PULSES FOUR	ROUTINING FAULT
18	FORCE SECTOR MARKER	BUFFER PARITY FAULT
19	FORCE INDEX MARKER	DISC NOT ROTATING
20	FORCE POWER FAULT	POWER FAULT
21	STOP	STOPPED
22	FREEZE FAULT RESET	FROZEN
23	SET OUT OF SERVICE	OUT OF SERVICE

CONSOLE CONTROL UNIT REGISTERS (1)

BIT	COMMAND	OPERATIONAL STATUS
0	ERASE SECTOR	SPARE
1	ERASE TO END OF TRACK	TRANSFER NOT COMPLETE
2	BACK SPACE	WRITE PROTECTED
3	REWIND	BEGINNING OF TRACK
4	WRITE/READ ENABLE	END OF TRACK
5	SPARE	DATA IN FULL
6	"	DATA OUT FULL
7	ECHO	ECHO
8	SPARE	SPARE
9	INHIBIT DATA CHECK	DATA CHECK INHIBITED
10	INVERT BUFFER PARITY	BUFFER PARITY INVERTED
11	INHIBIT FREEZE	FREEZE INHIBITED
12	SET INITIALISING	INITIALISING
13	SPARE	ACTIVE
14	"	SPARE
15	"	BUFFER PARITY FAULT
16	STOP CCU	CCU STOPPED
17	SPARE	SPARE
18	"	OVERRUN
19	"	VDU NOT READY
20	NON FREEZE FAULT RESET	TAPE UNIT NOT LOADED
21	FREEZE FAULT RESET	ERROR IN SECONDARY STATUS
22	SET OUT OF SERVICE VDU	VDU OUT OF SERVICE
23	SET OUT OF SERVICE TAPE	TAPE OUT OF SERVICE

CONSOLE CONTROL UNIT REGISTERS (2)

BIT	SECONDARY STATUS	DIAGNOSTIC
0	ERASING SECTOR	DATA TAPE LOOP 2
1	ERASING TO END OF TRACK	SELECT 150 KHZ
2	BACK SPACING	SPARE
3	REWINDING	"
4	WRITE/READ ENABLE	FORCE READ ENABLE
5	SPARE	FORCE TAPE READY
6	"	SPARE
7	REWIND PULSE	FORCE DISPLAY READY
8	RUN	SPARE
9	REVERSE	"
10	WRITE/READ	"
11	TRACK ADDRESS 2^2	"
12	TRACK ADDRESS 2^1	FORCE TAPE NOT LOADED
13	TRACK ADDRESS 2^0	FORCE END OF TRACK
14	SPARE	FORCE BEGINNING OF TRACK
15	"	FORCE CLOCK 2^7
16	"	FORCE CLOCK 2^6
17	TRANSFER TIMEOUT	FORCE CLOCK 2^5
18	DATA CHECK FAULT	FORCE CLOCK 2^4
19	WORDS PER SECTOR FAULT	FORCE CLOCK 2^3
20	DATA BUFFER OVERFLOW	FORCE CLOCK 2^2
21	INTERNAL PARITY FAULT	FORCE CLOCK 2^1
22	WRITE PARITY FAULT	FORCE CLOCK 2^0
23	POWER FAULT	FORCE POWER FAULT

EXCHANGEABLE DISC SYSTEM REGISTERS (1)

BIT	COMMAND	STATUS
0	SPARE	DRIVE FAULT
1	"	OFF CYLINDER
2	"	SEEK ERROR
3	"	UNIT NOT READY
4	"	SPARE
5	"	"
6	"	"
7	"	"
8	"	"
9	"	"
10	"	"
11	"	"
12	INHIBIT DATA CHECK	DATA CHECK INHIBITED
13	INVERT PARITY	PARITY INVERTED
14	FORCE FREEZE ON ALL FAULTS	FORCE FREEZE ON ALL FAULTS SET
15	SPARE	SPARE
16	"	INVALID DRIVE UNIT SELECTED
17	"	INVALID WORK ITEM
18	"	BUFFER PARITY FAULT
19	"	DATA BUFFER OVERFLOW
20	FORCE POWER FAULT	POWER FAULT
21	STOP	STOPPED
22	FAULT RESET	FROZEN
23	SET OUT OF SERVICE	OUT OF SERVICE

EXCHANGEABLE DISC SYSTEM REGISTERS (2)

BIT	WORK ITEM	WORK STATUS
0		DRIVE FAULT
1		OFF CYLINDER
2		SEEK ERROR
3	ADDRESS FIELD: FOR READ, WRITE AND FORMAT THIS FIELD SPECIFIES ADDRESS IN BUFFER OF TRANSFER CAPABILITY. FOR SEEK THIS FIELD SPECIFIES ADDRESS TO BE SOUGHT.	UNIT NOT READY
4		TRANSFER CAPABILITY FAULT
5		CRC FAULT ON DATA
6		SECTOR ADDRESS FAULT
7		INVALID SEEK
8		TRANSFER TIMEOUT
9		WRITE PARITY FAULT
10		INTERNAL PARITY FAULT
11		
12	DISC DRIVE UNIT SELECT	DISC DRIVE UNIT SELECTED
13		
14		
15	PAUSE	PAUSE
16	MONITOR DRIVE UNIT { DATA STROBE LATE * / DATA STROBE EARLY * }	MONITOR DRIVE UNIT { DATA STROBE LATE * / DATA STROBE EARLY * }
17		
18	FAULT CLEAR	FAULT CLEAR
19	RETURN TO TRACK ZERO { SERVO OFFSET - / SERVO OFFSET + }	RETURN TO TRACK ZERO { SERVO OFFSET - / SERVO OFFSET + }
20	FORMAT	FORMAT
21	SEEK *	SEEK
22	READ	READ
23	WRITE	WRITE

* ONLY APPLY WHEN READ IS SELECTED

EXCHANGEABLE DISC SYSTEM DIAGNOSTIC REGISTERS

BIT	DIAGNOSTIC 1	DIAGNOSTIC 2	DIAGNOSTIC 3	
0	BIT 0 DATA	FORCE DRIVE FAULT	SELECT DRIVE 1	STAR SELECT BOARD 1
1	BIT 1 DATA	FORCE OFF CYLINDER	" " 2	
2	BIT 2 DATA	FORCE SEEK ERROR	" " 3	
3	BIT 3 DATA	FORCE UNIT NOT READY	" " 4	
4	BIT 4 DATA	FORCE INDEX MARKER	" " 5	STAR SELECT BOARD 2
5	BIT 5 DATA	FORCE SECTOR MARKER	" " 6	
6	BIT 6 DATA	1 OR 24 PULSES ON READ CLOCK *	" " 7	
7	BIT 7 DATA	CLOCK PROGRAM CLOCK	" " 8	
8	BIT 8 DATA	CLOCK READ CLOCK	" " 9	
9	BIT 9 DATA	CLOCK SERVO CLOCK	" " 10	STAR SELECT BOARD 3
10	SPARE	SERIAL DATA SWITCH IN	" " 11	
11	UNIT SELECT TAG	SERIAL DATA IN	" " 12	
12	UNIT SELECT 2^0		" " 13	STAR SELECT BOARD 4
13	UNIT SELECT 2^1		" " 14	
14	UNIT SELECT 2^2		" " 15	
15	UNIT SELECT 2^3		NOT USED	
16	TAG 1			
17	TAG 2			
18	TAG 3			
19	SERIAL DATA BIT			
20	SPARE			
21	"			
22	"			
23	"			

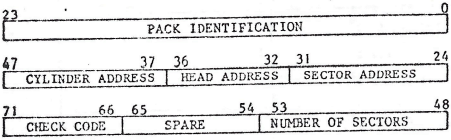
* 1 FOR 24, 0 FOR 1.

OUTPUTS ONLY FORCED WHEN STOP BIT IS SET.

BUS LINE SPECIFICATION FOR CDC DRIVE UNIT

BIT	TAG 1 CYLINDER ADDRESS	TAG 2 HEAD SELECT	TAG 3 CONTROL SELECT
0	2^0	2^0	WRITE GATE
1	2^1	2^1	READ GATE
2	2^2	2^2	SERVO OFFSET +
3	2^3	-	SERVO OFFSET -
4	2^4	-	FAULT CLEAR
5	2^5	-	-
6	2^6	-	RETURN TO TRACK ZERO
7	2^7	-	DATA STROBE EARLY
8	2^8	-	DATA STROBE LATE

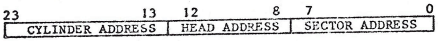
THE TRANSFER CAPABILITY



TRANSFER CAPABILITY CHECK CODE

A = 1 @ 7 @ 13 @ 19 @ 27 @ 33 @ 39 @ 45 @ 53 @ 59 @ 65
B = 0 @ 6 @ 12 @ 18 @ 26 @ 32 @ 38 @ 44 @ 52 @ 58 @ 64
C = 5 @11 @ 17 @ 23 @ 25 @ 31 @ 37 @ 43 @ 51 @ 57 @ 63
D = 4 @10 @ 16 @ 22 @ 24 @ 30 @ 36 @ 42 @ 50 @ 56 @ 62
E = 3 @ 9 @ 15 @ 21 @ 29 @ 35 @ 41 @ 47 @ 49 @ 55 @ 61
F = 2 @ 8 @ 14 @ 20 @ 28 @ 32 @ 40 @ 46 @ 48 @ 54 @ 60

LAST SECTOR ADDRESS



CHARACTER CODE

Char.	8 bit ASCII	Hex. (No parity bit)
A	101	41
B	102	42
C	303	43
D	104	44
E	305	45
F	306	46
G	107	47
H	110	48
I	311	49
J	312	4A
K	113	4B
L	314	4C
M	115	4D
N	116	4E
O	317	4F
P	120	50
Q	321	51
R	322	52
S	123	53
T	324	54
U	125	55
V	126	56
W	327	57
X	330	58
Y	131	59
Z	132	5A
Space	240	20
Car. Ret.	215	0D
Line Feed	012	0A
Bell	207	87
@	300	40
&	246	3F

Char.	8 bit ASCII	Hex. (No parity bit)
∅	060	30
1	261	31
2	262	32
3	063	33
4	264	34
5	065	35
6	066	36
7	267	37
8	270	38
9	071	39
[	333	5B
\	234	5C
]	335	5D
:	072	3A
;	273	3B
<	074	3C
=	275	3D
>	276	3E
?	077	3F
!	041	21
"	042	22
#	243	23
¤	245	25
'	047	27
(	050	28
)	251	29
*	252	2A
+	053	2B
,	254	2C
-	055	2D
.	056	2E
/	257	2F

POWERS OF TWO

0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1 024
11	2 048

12	4 096
13	8 192
14	16 384
15	32 764
16	65 536
17	131 072
18	262 144
19	524 288
20	1 048 576
21	2 096 152
22	4 194 304
23	8 388 608

OCTAL(*) - DECIMAL CONVERSION TABLE

*		*		*		*		*		*		*	
0000	0	0106	70	0214	140	0322	210	0430	280	0536	350	0644	420
0001	1	0107	71	0215	141	0323	211	0431	281	0537	351	0645	421
0002	2	0110	72	0216	142	0324	212	0432	282	0540	352	0646	422
0003	3	0111	73	0217	143	0325	213	0433	283	0541	353	0647	423
0004	4	0112	74	0220	144	0326	214	0434	284	0542	354	0650	424
0005	5	0113	75	0221	145	0327	215	0435	285	0543	355	0651	425
0006	6	0114	76	0222	146	0330	216	0436	286	0544	356	0652	426
0007	7	0115	77	0223	147	0331	217	0437	287	0545	357	0653	427
0010	8	0116	78	0224	148	0332	218	0440	288	0546	358	0654	428
0011	9	0117	79	0225	149	0333	219	0441	289	0547	359	0655	429
0012	10	0120	80	0226	150	0334	220	0442	290	0550	360	0656	430
0013	11	0121	81	0227	151	0335	221	0443	291	0551	361	0657	431
0014	12	0122	82	0230	152	0336	222	0444	292	0552	362	0660	432
0015	13	0123	83	0231	153	0337	223	0445	293	0553	363	0661	433
0016	14	0124	84	0232	154	0340	224	0446	294	0554	364	0662	434
0017	15	0125	85	0233	155	0341	225	0447	295	0555	365	0663	435
0020	16	0126	86	0234	156	0342	226	0450	296	0556	366	0664	436
0021	17	0127	87	0235	157	0343	227	0451	297	0557	367	0665	437
0022	18	0130	88	0236	158	0344	228	0452	298	0560	368	0666	438
0023	19	0131	89	0237	159	0345	229	0453	299	0561	369	0667	439
0024	20	0132	90	0240	160	0346	230	0454	300	0562	370	0670	440
0025	21	0133	91	0241	161	0347	231	0455	301	0563	371	0671	441
0026	22	0134	92	0242	162	0350	232	0456	302	0564	372	0672	442
0027	23	0135	93	0243	163	0351	233	0457	303	0565	373	0673	443
0030	24	0136	94	0244	164	0352	234	0460	304	0566	374	0674	444
0031	25	0137	95	0245	165	0353	235	0461	305	0567	375	0675	445
0032	26	0140	96	0246	166	0354	236	0462	306	0570	376	0676	446
0033	27	0141	97	0247	167	0355	237	0463	307	0571	377	0677	447
0034	28	0142	98	0250	168	0356	238	0464	308	0572	378	0700	448
0035	29	0143	99	0251	169	0357	239	0465	309	0573	379	0701	449
0036	30	0144	100	0252	170	0360	240	0466	310	0574	380	0702	450
0037	31	0145	101	0253	171	0361	241	0467	311	0575	381	0703	451
0040	32	0146	102	0254	172	0362	242	0470	312	0576	382	0704	452
0041	33	0147	103	0255	173	0363	243	0471	313	0577	383	0705	453
0042	34	0150	104	0256	174	0364	244	0472	314	0600	384	0706	454
0043	35	0151	105	0257	175	0365	245	0473	315	0601	385	0707	455
0044	36	0152	106	0260	176	0366	246	0474	316	0602	386	0710	456
0045	37	0153	107	0261	177	0367	247	0475	317	0603	387	0711	457
0046	38	0154	108	0262	178	0370	248	0476	318	0604	388	0712	458
0047	39	0155	109	0263	179	0371	249	0477	319	0605	389	0713	459
0050	40	0156	110	0264	180	0372	250	0500	320	0606	390	0714	460
0051	41	0157	111	0265	181	0373	251	0501	321	0607	391	0715	461
0052	42	0160	112	0266	182	0374	252	0502	322	0610	392	0716	462
0053	43	0161	113	0267	183	0375	253	0503	323	0611	393	0717	463
0054	44	0162	114	0270	184	0376	254	0504	324	0612	394	0720	464
0055	45	0163	115	0271	185	0377	255	0505	325	0613	395	0721	465
0056	46	0164	116	0272	186	0400	256	0506	326	0614	396	0722	466
0057	47	0165	117	0273	187	0401	257	0507	327	0615	397	0723	467
0060	48	0166	118	0274	188	0402	258	0510	328	0616	398	0724	468
0061	49	0167	119	0275	189	0403	259	0511	329	0617	399	0725	469
0062	50	0170	120	0276	190	0404	260	0512	330	0620	400	0726	470
0063	51	0171	121	0277	191	0405	261	0513	331	0621	401	0727	471
0064	52	0172	122	0300	192	0406	262	0514	332	0622	402	0730	472
0065	53	0173	123	0301	193	0407	263	0515	333	0623	403	0731	473
0066	54	0174	124	0302	194	0410	264	0516	334	0624	404	0732	474
0067	55	0175	125	0303	195	0411	265	0517	335	0625	405	0733	475
0070	56	0176	126	0304	196	0412	266	0520	336	0626	406	0734	476
0071	57	0177	127	0305	197	0413	267	0521	337	0627	407	0735	477
0072	58	0200	128	0306	198	0414	268	0522	338	0630	408	0736	478
0073	59	0201	129	0307	199	0415	269	0523	339	0631	409	0737	479
0074	60	0202	130	0310	200	0416	270	0524	340	0632	410	0740	480
0075	61	0203	131	0311	201	0417	271	0525	341	0633	411	0741	481
0076	62	0204	132	0312	202	0420	272	0526	342	0634	412	0742	482
0077	63	0205	133	0313	203	0421	273	0527	343	0635	413	0743	483
0100	64	0206	134	0314	204	0422	274	0530	344	0636	414	0744	484
0101	65	0207	135	0315	205	0423	275	0531	345	0637	415	0745	485
0102	66	0210	136	0316	206	0424	276	0532	346	0640	416	0746	486
0103	67	0211	137	0317	207	0425	277	0533	347	0641	417	0747	487
0104	68	0212	138	0320	208	0426	278	0534	348	0642	418	0750	488
0105	69	0213	139	0321	209	0427	279	0535	349	0643	419	0751	489